Magnetization scaling of Fortuin–Kasteleyn clusters in the two-dimensional $\pm J$ random-bond Ising model

Abstract

We define an exponent θ through $\langle |M_C| \rangle \sim \langle S \rangle^{\theta}$, relating the net magnetization M_C carried by the largest Fortuin–Kasteleyn cluster to its size S. We show that a Fortuin–Kasteleyn cluster of the $\pm J$ model admits a well-defined gauge, that the gauge is exponentially correlated along paths with a length $\xi_{\mathrm{g}} = -1/\ln(1-2p)$ set by the disorder alone, and that consequently $\langle M_C^2 \rangle \propto S$ whenever the cluster is much larger than ξ_{g}. Hence $\theta = 1/2$, with an amplitude proportional to ξ_{g}. The complementary limit $\theta = 1$ holds identically in the pure ferromagnet at every temperature, so any departure from unity is purely a disorder effect. We state the one place where the argument is heuristic rather than rigorous, and compare with simulation.

1 Setting and definitions

Consider the $\pm J$ random-bond Ising model on a finite lattice,

$$H = -\sum_{\langle ij \rangle} J_{ij}\, s_i s_j, \qquad s_i = \pm 1, \qquad J_{ij} = \begin{cases} -1 & \text{with probability } p, \\ +1 & \text{with probability } 1-p, \end{cases} \tag{1}$$

the couplings being independent across bonds. A bond $\langle ij \rangle$ is called *satisfied* in a configuration $\{s\}$ when $J_{ij} s_i s_j > 0$. In the Fortuin–Kasteleyn–Coniglio–Klein construction only satisfied bonds may be activated, each independently with probability $1 - e^{-2\beta}$; the clusters are the connected components of the active bonds. Throughout, C denotes such a cluster, $S = |C|$ its number of sites, and

$$M_C \;=\; \sum_{i \in C} s_i \tag{2}$$

the net magnetization it carries. The exponent of interest is defined by

$$\big\langle |M_C| \big\rangle \;\sim\; \big\langle S \big\rangle^{\theta}, \tag{3}$$

the averages being over Monte Carlo time and disorder at fixed (p, T), and the scaling taken with respect to linear system size. Two values are distinguished: $\theta = 1$ means every spin of the cluster contributes coherently, and $\theta = 1/2$ means the spins add as a random walk, so that the cluster carries no coherent moment.

2 The cluster carries a gauge

Lemma 1 (No frustrated loops)**.** *Let γ be any closed loop of bonds contained in a cluster C. Then* $\prod_{\langle ij \rangle \in \gamma} \operatorname{sgn}(J_{ij}) = +1$.

Proof. Every bond of C is active, hence satisfied, so $s_i s_j = \operatorname{sgn}(J_{ij})$ on each. Taking the product around γ, each site appears exactly twice, so $\prod_{\gamma} s_i s_j = +1$, and therefore $\prod_{\gamma} \operatorname{sgn}(J_{ij}) = +1$. □

Lemma 2 (Gauge representation)**.** *Fix a reference site $o \in C$. For $i \in C$ let P be any path from o to i lying in C, and set*

$$\sigma_i \;=\; \prod_{\langle kl\rangle \in P} \operatorname{sgn}(J_{kl}). \tag{4}$$

Then σ_i does not depend on the choice of P, and $s_i = s_o\,\sigma_i$ for every $i \in C$.

Proof. Two paths from o to i differ by a closed loop in C, which contributes $+1$ by Lemma 1; hence σ_i is well defined. The identity $s_i = s_o\sigma_i$ follows by induction along P, using $s_k = s_l \operatorname{sgn}(J_{kl})$ on each satisfied bond. □

Corollary 1. *$M_C = s_o \sum_{i\in C} \sigma_i$, and therefore $|M_C| = \bigl|\sum_{i\in C} \sigma_i\bigr|$.*

The content of Corollary 1 is that the cluster is always internally ordered: its spins are rigidly locked to one another up to the single global sign s_o. What varies is the *gauge pattern* $\{\sigma_i\}$, which is a property of the disorder and of the cluster geometry, and not of the temperature. The magnetization of the cluster is the projection of that pattern onto the uniform direction.

3 The gauge is exponentially correlated

Lemma 3 (Two-point function)**.** *Let $i, j \in C$ and let ℓ_{ij} be the number of bonds on a path joining them within C. Treating the couplings as independent,*

$$\mathbb{E}\bigl[\sigma_i\sigma_j\bigr] \;=\; (1-2p)^{\ell_{ij}} \;=\; e^{-\ell_{ij}/\xi_{\mathrm{g}}}, \qquad \boxed{\xi_{\mathrm{g}} \equiv -\frac{1}{\ln(1-2p)}} \tag{5}$$

Proof. By Lemma 2, $\sigma_i\sigma_j$ equals the product of $\operatorname{sgn}(J)$ over the bonds of the path joining i to j, each bond occurring once. For a single bond $\mathbb{E}[\operatorname{sgn}(J)] = (1-p) - p = 1-2p$, and independence factorizes the product over the ℓ_{ij} distinct bonds. □

The length ξ_{g} is fixed by the disorder concentration alone. It is finite for every $p > 0$ and diverges as $p \to 0$.

4 Main result

Proposition 1 ($\theta = 1/2$)**.** *Suppose the number of sites of C at chemical distance ℓ from a given site grows at most polynomially, $N(\ell) \le c\,\ell^{d_\ell - 1}$ for some d_ℓ and constant c. Then*

$$\mathbb{E}\bigl[M_C^2\bigr] \;=\; n_\infty\, S\,\bigl(1 + o(1)\bigr), \qquad n_\infty \;=\; \sum_{\ell \ge 0} N(\ell)\, e^{-\ell/\xi_{\mathrm{g}}} \;<\; \infty, \tag{6}$$

where n_∞ depends on ξ_{g} and on the local geometry of the cluster but not on S. Consequently $|M_C| \sim \sqrt{n_\infty S}$ and

$$\theta = \tfrac{1}{2}. \tag{7}$$

Proof. By Corollary 1 and Lemma 3,

$$\mathbb{E}\bigl[M_C^2\bigr] = \sum_{i,j\in C} \mathbb{E}[\sigma_i\sigma_j] = \sum_{i\in C} \underbrace{\sum_{j\in C} e^{-\ell_{ij}/\xi_{\mathrm{g}}}}_{n_i}. \tag{8}$$

Grouping the inner sum by chemical distance gives $n_i = \sum_\ell N_i(\ell)\, e^{-\ell/\xi_{\mathrm{g}}}$, and under the stated growth condition

$$n_i \;\le\; c \sum_{\ell \ge 0} \ell^{\,d_\ell - 1} e^{-\ell/\xi_{\mathrm{g}}} \;<\; \infty, \tag{9}$$

since an exponential dominates any power. The bound is uniform in S, so n_i saturates at a finite value n_∞ once the cluster's chemical diameter exceeds a few ξ_{g}. Hence $\mathbb{E}[M_C^2] = n_\infty S$, and the root-mean-square magnetization grows as $\sqrt{S}$. Comparing with (3) gives $\theta = 1/2$. □

Corollary 2 (Amplitude)**.** *Writing* $\langle |M_C| \rangle = A\sqrt{\langle S \rangle}$ *and using* $\langle |X| \rangle = \sqrt{2/\pi}\, \langle X^2 \rangle^{1/2}$ *for an approximately Gaussian* M_C*,*

$$A \;=\; \sqrt{\tfrac{2}{\pi} n_\infty} \;\propto\; \xi_{\mathrm{g}}. \tag{10}$$

For a compact two-dimensional cluster with chemical distance comparable to Euclidean distance, $N(\ell) \simeq 2\pi\ell$ *and* $n_\infty \simeq 2\pi\xi_{\mathrm{g}}^2$*, giving* $A \simeq 2\,\xi_{\mathrm{g}}$*.*

Remark 1 (Regime of validity). Proposition 1 requires the cluster to be much larger than the gauge correlation length. When instead $S^{1/d} \ll \xi_{\mathrm{g}}$, every site of the cluster is coherent, $M_C = \pm S$, and $\theta = 1$. The exponent therefore interpolates between 1 and 1/2 according to cluster size relative to ξ_{g}, and is not by itself a statement about the thermodynamic phase. What links it to the physics is that ferromagnetic order requires the *spanning* cluster to be gauge-coherent: for the spanning cluster, $\theta = 1$ is order and $\theta = 1/2$ is its absence.

Remark 2 (The pure ferromagnet). At $p = 0$ a satisfied bond joins parallel spins, so every Fortuin–Kasteleyn cluster consists of aligned spins and $M_C = \pm S$ identically. Thus $\theta \equiv 1$ at *every* temperature, including deep in the paramagnetic phase. This is the Coniglio–Klein identity $m = P_\infty$ in the present language, and it shows that any departure of θ from unity is a disorder effect rather than a thermal one.

Remark 3 (Where the argument is heuristic). Lemma 3 treats the couplings along the path as unconditioned. In fact the cluster C is determined jointly by the disorder and the spin configuration, so conditioning on $i, j \in C$ biases the law of the intervening bonds, and $\mathbb{E}[\sigma_i\sigma_j \mid i, j \in C] \neq (1-2p)^{\ell_{ij}}$ exactly. The exponential decay structure, and with it $\theta = 1/2$, is expected to survive this conditioning; the constant of proportionality in Corollary 2 is not controlled by the present argument. The numerical evidence below bears on precisely this point.

5 Comparison with simulation

Swendsen–Wang simulations of the two-dimensional $\pm J$ model, with 10^4 disorder realizations at each of $L = 16, 24, 32, 48$ and 64, give the following at the Fortuin–Kasteleyn percolation temperature of each disorder concentration. Errors on θ combine a parametric bootstrap with a systematic term from the drift between consecutive-size slopes; the systematic dominates.

p	θ	$A = \langle \lvert M_C \rvert \rangle / \sqrt{\langle S \rangle}$	ξ_{g}	A/ξ_{g}
0.02	0.940 ± 0.020	21.08	24.5	—
0.05	0.672 ± 0.086	13.77	9.49	—
0.08	0.513 ± 0.003	6.87	5.74	1.20
0.10	0.502 ± 0.008	5.07	4.48	1.13
0.15	0.499 ± 0.004	3.11	2.80	1.11
0.20	0.4998 ± 0.0008	2.19	1.96	1.12

For the three concentrations at which θ has converged to $1/2$, the ratio A/ξ_{g} is constant to two percent across a factor of 2.3 in ξ_{g}, confirming Corollary 2 and indicating that the conditioning discussed above does not alter the scaling. The measured constant, $A \simeq 1.12\,\xi_{\mathrm{g}}$, is smaller than the compact-cluster estimate $A \simeq 2\xi_{\mathrm{g}}$, as expected: chemical distance inside a diluted Fortuin–Kasteleyn cluster exceeds Euclidean distance, which accelerates the decay in (5). At $p = 0.02$ and $p = 0.05$ the gauge length is comparable to the largest lattices, the cluster is not in the regime $S^{1/d} \gg \xi_{\mathrm{g}}$, and neither $\theta = 1/2$ nor the amplitude law is expected to hold; both entries behave accordingly.

6 Relation to earlier work

The $\sqrt{S}$ behaviour itself is not new. In the symmetric two-dimensional Ising spin glass the magnetization of the largest cluster divided by the square root of its volume is known to be insensitive to system size, a result usually motivated by the droplet picture: a compact cluster contains $V \sim L^2$ spins and acquires a magnetization $\sim \sqrt{V}$ from randomness [1]. What the present derivation adds is the identification of the controlling length as the gauge correlation length $\xi_{\mathrm{g}} = -1/\ln(1-2p)$, the resulting crossover from $\theta = 1$ to $\theta = 1/2$ as a function of disorder, and the parameter-free prediction $A \propto \xi_{\mathrm{g}}$ that the simulation confirms.

References

[1] Magnetic exponents of two-dimensional Ising spin glasses, arXiv:0704.1626.